Introduction to Analysis Homework 6

April, 10, 2025

1. (8 pt) Let $f : \mathbb{R}^3 \to \mathbb{R}$ be C^2 and let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be C^2. Then show that
   (a) $curl(grad(f)) = \mathbf{0}$.
   (b) $div(curl(\mathbf{F})) = 0$.
2. (12 pt) Prove that $\int_1^\infty \frac{\sin x}{x} dx$ converges but is not absolutely convergent.
3. (10 pt) Let $D = [0, 1]^2 \subset \mathbb{R}^2$. Calculate $\oint_{\partial D} \mathbf{f}(\mathbf{r}) \cdot d\mathbf{r}$, where $\mathbf{f} = (6y^2, 2x - 2y^4)$, and ∂D is the boundary of D in the positive direction.
4. (10 pt) Find the area of D bounded by the curve parameterized by $(\sin(2t), \sin(t))$ for $t \in [0, \pi]$.
5. (10 pt) Compute $\iint_{\partial E} \mathbf{f} \cdot \mathbf{n} dS$, where $\mathbf{f} = (3x + z^{2025}, y^2 - \cos^2(x), xz + x^{y^6})$ and ∂E is the surface of box

   $$0 \leq x \leq 1,\ 0 \leq y \leq 3,\ 0 \leq z \leq 2,$$

   use outward normal $\mathbf{n}$.